Hadoop & Spark - ISEN 2025/2026

Lab: Actions & Transformations

For this Lab you will need:

- The 3.3 Lab

As a reminder

Check 3.1 to login in NGROK, Kaggle, install Java or Spark and import libraries.

Transformations & Actions

Lets see a concrete example of Transformations vs Actions.

Transformations & Actions

First take a look at your Spark UI and check what was the last task/job executed.

Spark Jobs (?)

User: root
Total Uptime: 1.3 h
Scheduling Mode: FIFO
Completed Jobs: 109
Failed Jobs: 5

Event Timeline

Completed Jobs (109)

Page: 1 2 >

2 Pages. Jump to 1 . Show 100 items in a page. Go

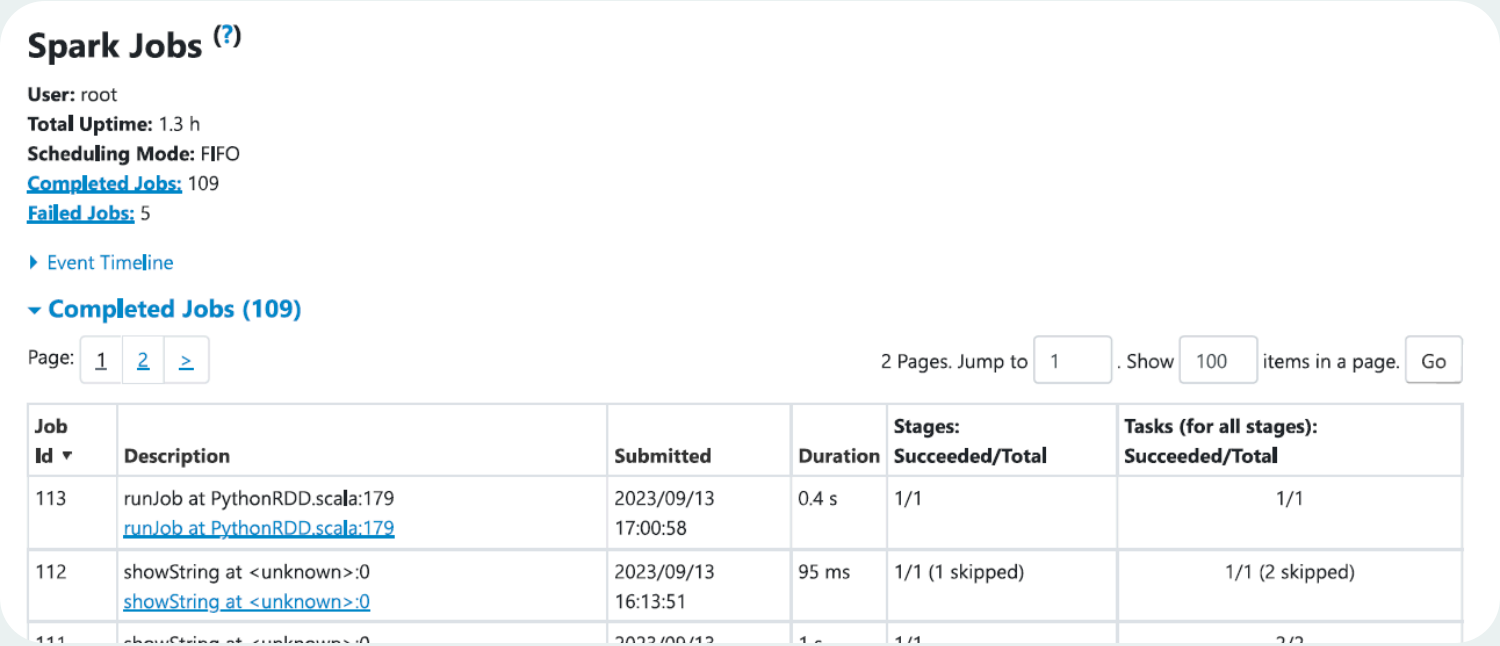
Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
113	runJob at PythonRDD.scala:179 runJob at PythonRDD.scala:179	2023/09/13 17:00:58	0.4 s	1/1	1/1
112	showString at <unknown>:0 showString at <unknown>:0	2023/09/13 16:13:51	95 ms	1/1 (1 skipped)	1/1 (2 skipped)
111	showString at <unknown>:0	2023/09/13	1 s	1/1	2/2

Transformations & Actions

Then launch this line:

```
RDD = sc.parallelize([1,2,3,4,5,6]) # Transformation
```

As this is a Transformation and not an Action, nothing happened and you can see no changes in the UI.



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Page: 1 2 > 2 Pages. Jump to 1 . Show 100 items in a page. Go

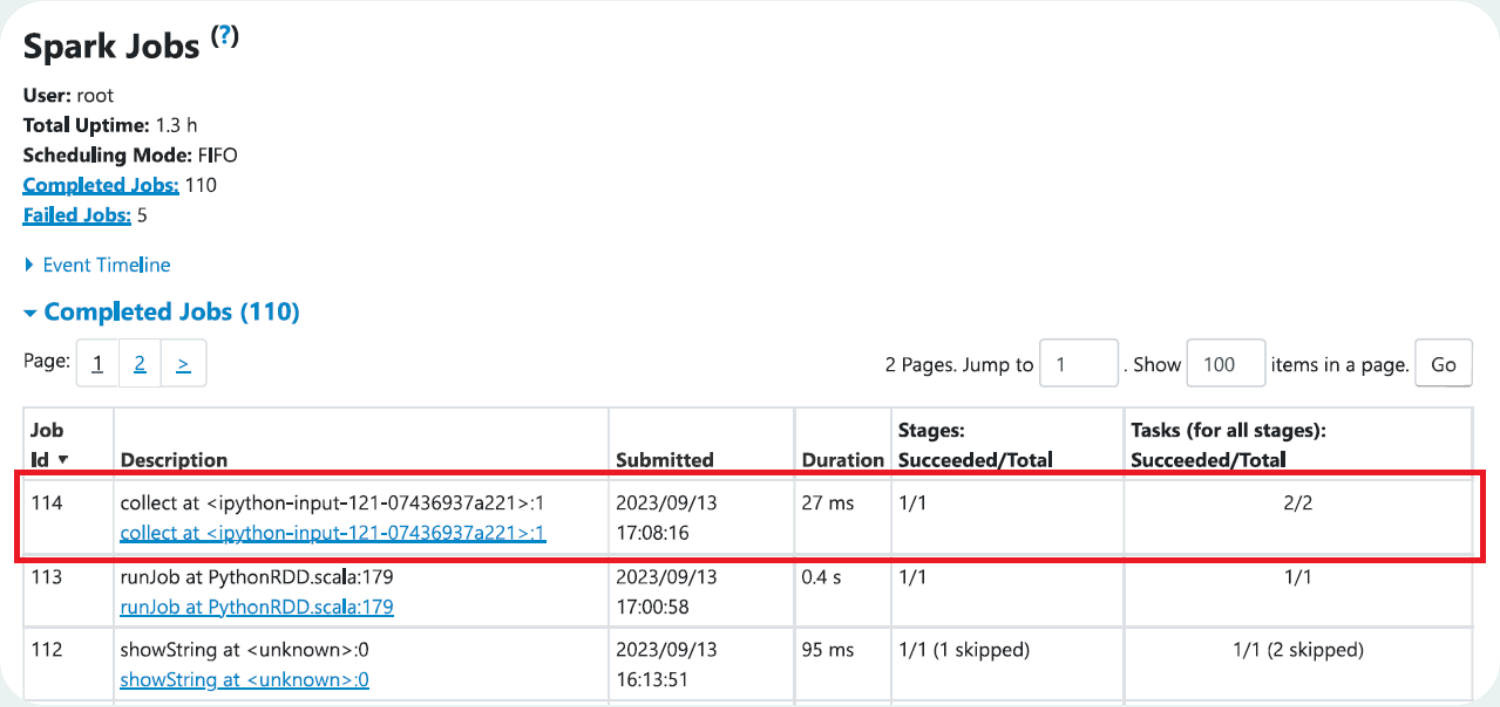
Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
113	runJob at PythonRDD.scala:179 runJob at PythonRDD.scala:179	2023/09/13 17:00:58	0.4 s	1/1	1/1
112	showString at <unknown>:0 showString at <unknown>:0	2023/09/13 16:13:51	95 ms	1/1 (1 skipped)	1/1 (2 skipped)

Transformations & Actions

Now let's launch an Action:

```
RDD.collect() # Action
```

If you check the UI, you can now see the job.



Spark Jobs (?)

User: root
Total Uptime: 1.3 h
Scheduling Mode: FIFO
Completed Jobs: 110
Failed Jobs: 5

Event Timeline

Completed Jobs (110)

Page: 1 2 > 2 Pages. Jump to 1 . Show 100 items in a page. Go

Job Id	Description	Submitted	Duration	Stages: Succeeded/Total	Tasks (for all stages): Succeeded/Total
114	collect at <ipython-input-121-07436937a221>:1 collect at <ipython-input-121-07436937a221>:1	2023/09/13 17:08:16	27 ms	1/1	2/2
113	runJob at PythonRDD.scala:179 runJob at PythonRDD.scala:179	2023/09/13 17:00:58	0.4 s	1/1	1/1
112	showString at <unknown>:0 showString at <unknown>:0	2023/09/13 16:13:51	95 ms	1/1 (1 skipped)	1/1 (2 skipped)

Job DAG

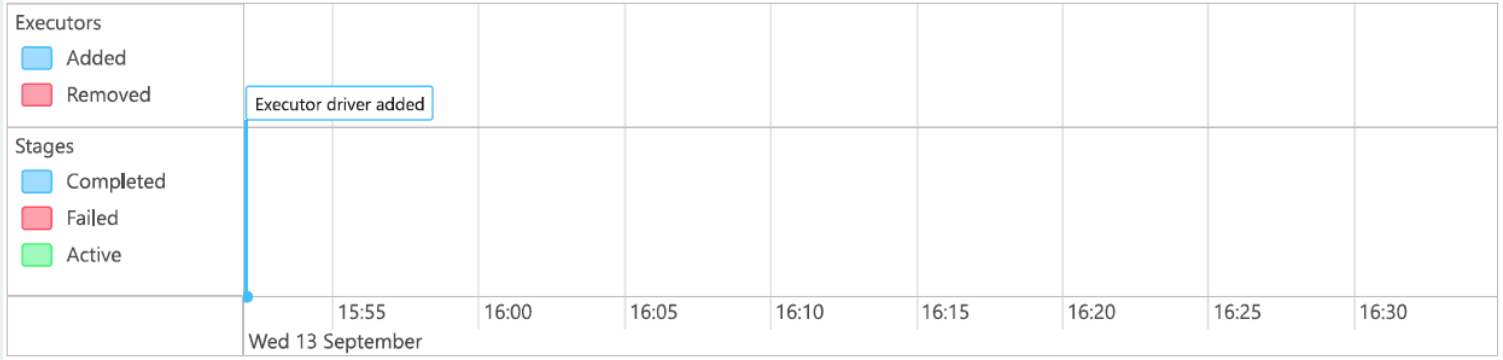
Now if you get back to the previous operations we made, you can see that Spark made a DAG every time we had operations going on.

Details for Job 114

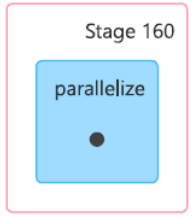
Status: SUCCEEDED
Submitted: 2023/09/13 17:08:16
Duration: 27 ms
Completed Stages: 1

Event Timeline

Enable zooming



DAG Visualization



Completed Stages (1)

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go

Stage Id	Description	Submitted	Duration	Tasks: Succeeded/Total	Input	Output	Shuffle Read	Shuffle Write
160	collect at <ipython-input-121-07436937a221>:1 +details	2023/09/13 17:08:16	20 ms	2/2				

Page: 1 — 1 Pages. Jump to 1 . Show 100 items in a page. Go